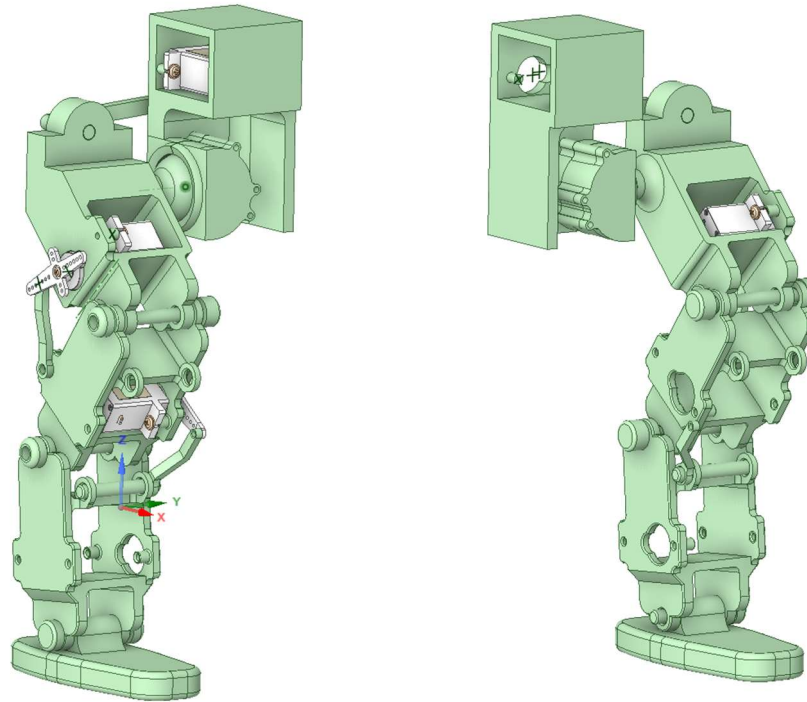


Design of a 3D-Printed Biped Robot



Project Overview

Objective:

To design and build a small, functional bipedal robot using 3D-printed parts, basic electronics, and servo motors. The robot will serve as an introduction to robotics as a proof of concept. It can potentially be taken up as research into the applications of robotics in the industrial and medical fields, such as prosthetics, service machinery, and factory assembly lines.

Scope

- Design : CAD modelling of biped robot parts (legs, torso, joints).
- 3D Printing : Fabricate structural components using available printers.
- Electronics : Servo motors (4-6 units), microcontroller (Arduino/Raspberry Pi), power supply, and wiring.
- Assembly & Testing: Integrate mechanical and electronic components, calibrate movement.

Required Resources

category	items	amount
3D printing	Filament (any available)	<300g
electronics	Servo motors, IMU unit	6, 1
tools	Soldering iron, screw drivers	

4. Expected Deliverables

- A functional bipedal robot capable of stable walking.
- Documentation (design files, assembly instructions, code).
- Lessons learned for future improvements (e.g., balance, gait efficiency).

Justification for the project

- Provision of skills development by providing Hands-on experience in robotics, 3D printing, and embedded systems.
- Cost-effective since printing reduces material costs vs. commercial kits.
- Provision for Scalability since it serves as a foundation for other projects like industrial assembly robots and the application of robotics in prosthetics.

Approval Request

Request access to:

- 3D printer & filament.
- Lab tools (if available).

3D PRINTING SPECIFICATIONS

Part group	Filament (grams)	Estimated time	Layer height
Left leg	122.42	8: 31	0.3mm
Right leg	122.42	8: 31	0.3mm
total	244.84	17: 02	

- Total grams: 244.84
- Total print time: 17 hours

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